PRACTICLE :- 11

Problem statement:-

Linux internally manages files using data structures known as inodes, which store metadata about files rather than file names. In addition, Linux provides powerful mechanisms such as input/output redirection, pipes, and process control commands to efficiently manage command execution and system resources. These concepts are fundamental for automation, scripting, system monitoring, and cyber security operations.

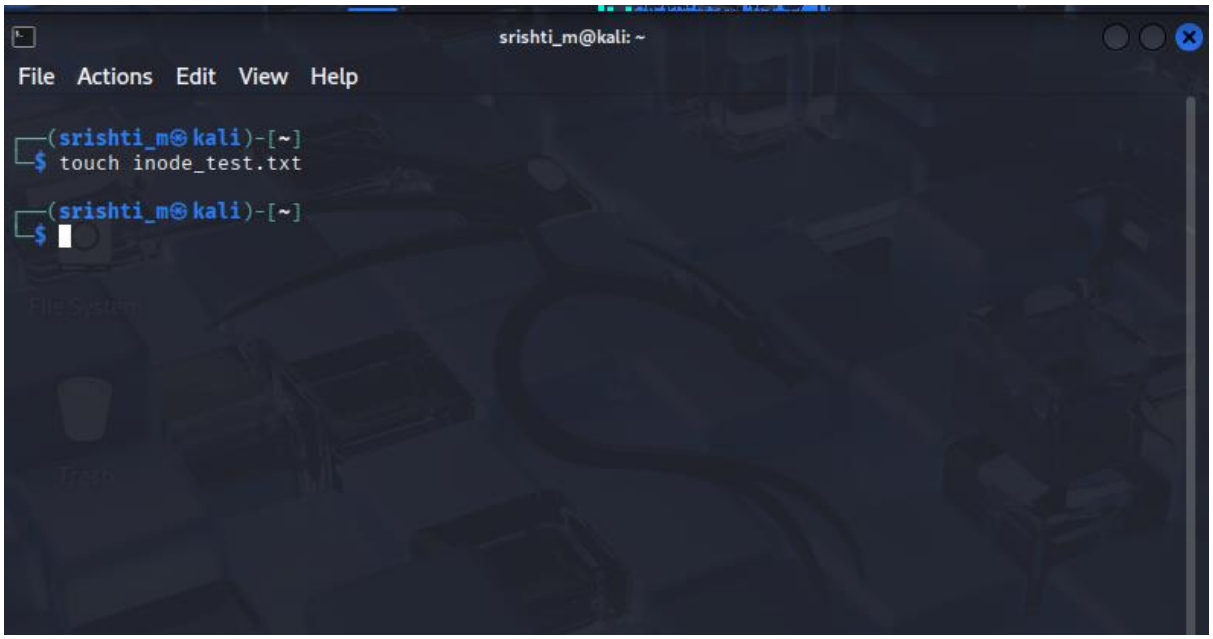
The problem is to understand the concept of inodes, practice I/O redirection and piping, and use process control commands to manage running processes in a Linux environment.

Aim :- Understanding inodes, I/O redirection, Piping and Process control commands in linux

Procedure:-

A. Understanding Inodes

1. Create a test file: touch inode_test.txt



2. Display inode number: ls -i inode_test.txt

```
File  Actions  Edit  View  Help

┌──(srishti_m㉿kali)-[~]
└─$ touch inode_test.txt

┌──(srishti_m㉿kali)-[~]
└─$ ls -i inode_test.txt
796674 inode_test.txt

┌──(srishti_m㉿kali)-[~]
└─$
```

3. View inode details: stat inode_test.txt

```
File  Actions  Edit  View  Help

┌──(srishti_m㉿kali)-[~]
└─$ touch inode_test.txt

┌──(srishti_m㉿kali)-[~]
└─$ ls -i inode_test.txt
796674 inode_test.txt

┌──(srishti_m㉿kali)-[~]
└─$ stat inode_test.txt
  File: inode_test.txt
  Size: 0               Blocks: 0          IO Block: 4096   regular empty file
Device: 8,1     Inode: 796674      Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/srishti_m)   Gid: ( 1000/srishti_m)
Access: 2026-03-17 02:08:31.338800631 -0400
Modify: 2026-03-17 02:08:31.338800631 -0400
Change: 2026-03-17 02:08:31.338800631 -0400
 Birth: 2026-03-17 02:07:28.783536811 -0400

┌──(srishti_m㉿kali)-[~]
└─$
```

B. Input / Output Redirection

4. Redirect output to a file: ls > file_list.txt

```
File  Actions  Edit  View  Help

┌──(srishti_m㉿kali)-[~]
└─$ touch inode_test.txt

┌──(srishti_m㉿kali)-[~]
└─$ ls -i inode_test.txt
796674 inode_test.txt

┌──(srishti_m㉿kali)-[~]
└─$ stat inode_test.txt
  File: inode_test.txt
  Size: 0               Blocks: 0          IO Block: 4096   regular empty file
Device: 8,1     Inode: 796674      Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/srishti_m)   Gid: ( 1000/srishti_m)
Access: 2026-03-17 02:08:31.338800631 -0400
Modify: 2026-03-17 02:08:31.338800631 -0400
Change: 2026-03-17 02:08:31.338800631 -0400
 Birth: 2026-03-17 02:07:28.783536811 -0400

┌──(srishti_m㉿kali)-[~]
└─$ ls >file_list.txt

┌──(srishti_m㉿kali)-[~]
└─$
```

5. Append output: date >> file_list.txt

```
┌──(srishti_m㉿kali)-[~]
└─$ date >> file_list.txt
```

6. Redirect input: wc -l < file_list.txt

```
796674 inode_test.txt

┌──(srishti_m㉿kali)-[~]
└─$ stat inode_test.txt
  File: inode_test.txt
  Size: 0               Blocks: 0          IO Block: 4096   regular empty file
Device: 8,1   Inode: 796674      Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/srishti_m)   Gid: ( 1000/srishti_m)
Access: 2026-03-17 02:08:31.338800631 -0400
Modify: 2026-03-17 02:08:31.338800631 -0400
Change: 2026-03-17 02:08:31.338800631 -0400
 Birth: 2026-03-17 02:07:28.783536811 -0400

┌──(srishti_m㉿kali)-[~]
└─$ ls >file_list.txt

┌──(srishti_m㉿kali)-[~]
└─$ date >> file_list.txt

┌──(srishti_m㉿kali)-[~]
└─$ ls > file_list.txt

┌──(srishti_m㉿kali)-[~]
└─$ date >> file_list.txt

┌──(srishti_m㉿kali)-[~]
└─$ wc -l< file_list.txt
21
```

C. Using Pipes

7. Combine commands using pipe: ls -l | grep ".txt"

```
┌──(srishti_m㉿kali)-[~]
└─$ ls -l |grep ".txt"
-rw-rw-r-- 1 srishti_m srishti_m       0 Feb 10 01:03 abc.txt
-rw-rw-r-- 1 srishti_m srishti_m     215 Mar 17 02:13 file_list.txt
-rw-rw-r-- 1 srishti_m srishti_m       0 Mar 17 02:08 inode_test.txt
```

8. Count number of text files: ls | grep ".txt" | wc -l

```
21

┌──(srishti_m㉿kali)-[~]
└─$ ls -l |grep ".txt"
-rw-rw-r-- 1 srishti_m srishti_m       0 Feb 10 01:03 abc.txt
-rw-rw-r-- 1 srishti_m srishti_m     215 Mar 17 02:13 file_list.txt
-rw-rw-r-- 1 srishti_m srishti_m       0 Mar 17 02:08 inode_test.txt

┌──(srishti_m㉿kali)-[~]
└─$ ls |grep".txt" | wc -l
Could not find command-not-found database. Run 'sudo apt update' to populate it.
grep.txt: command not found
0

┌──(srishti_m㉿kali)-[~]
└─$
```

D. Process Control Commands

9. Display running processes: ps

```
srishti_m@kali: ~
File  Actions  Edit  View  Help

┌──(srishti_m㉿kali)-[~]
└─$ ps
    PID TTY          TIME CMD
   7786 pts/0    00:00:00 bash
  14738 pts/0    00:00:00 ps

┌──(srishti_m㉿kali)-[~]
└─$
```

10. Monitor processes in real-time: top

```
top - 02:21:22 up 40 min,  1 user,  load average: 0.29, 0.34, 0.32
Tasks: 200 total,   1 running, 198 sleeping,   0 stopped,   1 zombie
%Cpu(s):  7.4 us,  3.8 sy,  0.0 ni, 88.7 id,  0.0 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem :   3686.4 total,   2522.4 free,    897.6 used,    486.0 buff/cache
MiB Swap:   3857.0 total,   3857.0 free,      0.0 used.   2788.8 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1189 srishti+  20   0    8740   4920   4280 S  22.9   0.1   2:43.70 dbus-daemon
   1343 srishti+  20   0  929476  85132  35924 S  21.6   2.3   2:41.56 orca
   1199 srishti+  20   0  168876   7400   6760 S  15.6   0.2   1:45.04 at-spi2-registr
    799 root      20   0  337260 103472  54996 S   4.7   2.7   1:03.97 Xorg
   1344 srishti+  20   0  587088  44312  35576 S   2.3   1.2   0:17.79 nm-applet
   1237 srishti+  20   0 1182860 125564  83096 S   1.7   3.3   0:19.38 xfwm4
   1295 srishti+  20   0  236568  60716  19820 S   0.7   1.6   0:12.13 wrapper-2.0
   7780 srishti+  20   0  719536  56980  48044 S   0.7   1.5   0:05.21 qterminal
    610 message+  20   0   10208   6268   4092 S   0.3   0.2   0:02.52 dbus-daemon
   1158 srishti+  20   0  215932   3100   2844 S   0.3   0.1   0:07.65 VBoxClient
   1296 srishti+  20   0  345988  25416  18800 S   0.3   0.7   0:04.40 wrapper-2.0
   1297 srishti+  20   0  274020  28452  21380 S   0.3   0.8   0:07.20 wrapper-2.0
   1341 root      20   0  318676  10216   8552 S   0.3   0.3   0:01.29 upowerd
  10452 srishti+  20   0  719536  56968  48148 S   0.3   1.5   0:00.81 qterminal
  14968 srishti+  20   0   10436   5632   3584 R   0.3   0.1   0:00.27 top
      1 root      20   0   23676  14340  10480 S   0.0   0.4   0:02.74 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.02 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_r+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
      9 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0-cgroup_des+
```

11. Run a command in background: sleep 60 &

```
┌──(srishti_m㉿kali)-[~]
└─$ sleep 60 &
[1] 15893

┌──(srishti_m㉿kali)-[~]
└─$
```

12. List background jobs: jobs

```
┌──(srishti_m㉿kali)-[~]
└─$ sleep 60 &
[1] 15893

┌──(srishti_m㉿kali)-[~]
└─$ jobs
[1]+  Running                 sleep 60 &

┌──(srishti_m㉿kali)-[~]
└─$
```

13. Bring job to foreground: fg

```
┌──(srishti_m㉿kali)-[~]
└─$ sleep 60 &
[1] 15893

┌──(srishti_m㉿kali)-[~]
└─$ jobs
[1]+  Running                 sleep 60 &

┌──(srishti_m㉿kali)-[~]
└─$ fg
sleep 60
```

14. Terminate a process: kill

```
┌──(srishti_m㉿kali)-[~]
└─$ sleep 60 &
[1] 15893

┌──(srishti_m㉿kali)-[~]
└─$ jobs
[1]+  Running                 sleep 60 &

┌──(srishti_m㉿kali)-[~]
└─$ fg
sleep 60

┌──(srishti_m㉿kali)-[~]
└─$ kill
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]

┌──(srishti_m㉿kali)-[~]
└─$
```

Expected Output

- Inode numbers and file metadata displayed correctly.
- Command output redirected and appended successfully.
- Pipes used to filter and process output.
- Processes monitored and controlled successfully.

Result

Inode information, input/output redirection, piping, and process control commands were successfully executed. The student gained practical understanding of Linux file system internals and process management.